

MathFoundations234:

**Alternating/ symmetric
polynumbers: a missing
chapter in Algebra**

copyright 2023 N J Wildberger

MF234: Alternating/symmetric polynomials- a missing chapter of Algebra

The subtraction operation If A, B are
msets ... or a/msets, then define

$$A + \bar{1}B = A + (-1)B$$

①

Ex

$$\begin{array}{|c|c|} \hline \square\square & \square\square \\ \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square\square & \square \\ \hline \square & \square \\ \hline \end{array} + \begin{array}{|c|} \hline \square \\ \hline \end{array} \times \begin{array}{|c|c|} \hline \square\square & \square\square \\ \hline \square & \square \\ \hline \end{array}$$

$$\begin{array}{|c|c|} \hline \square\square & \square \\ \hline \square & \square \\ \hline \end{array}$$

or $[1^a 0^a 2^a] - [2^a 2^a \bar{1}^a 0^a]$

$$= [\bar{1}^a 0^a 0^a 1^a 2^a]$$



Fundamental reflection symmetry σ

$$\boxed{\sigma(\alpha^n) = \alpha^{-n}} \quad \text{for } n \in \text{Int}$$

If $p \in \text{IPoly}$, then p is symmetric $\Leftrightarrow \boxed{\sigma(p) = p}$

while p is antisymmetric $\Leftrightarrow \boxed{\sigma(p) = -p = \bar{p}}$

②

Claim: The symmetric and antisymmetric polynomials, and their interaction, is a part of Algebra!

So, important families/types are A_n and $B_n, n \in \text{Int}$

Fundamental reflection symmetry σ

$$\boxed{\sigma(\alpha^n) = \alpha^{-n}} \quad \text{for } n \in \text{Int}$$

If $p \in \text{IPoly}$, then p is symmetric $\Leftrightarrow \boxed{\sigma(p) = p}$

p is alternating $\Leftrightarrow \boxed{\sigma(p) = -p = \bar{p}}$

②

The study of symmetric and alternating integral polys, and their interaction, is a missing chapter of Algebra!

let's introduce two important families/types of integral polynumbers: $A_n, n \in \text{Int}$ and $B_n, n \in \text{Int}$

Symmetric binomials $B_n, n \in \mathbb{Z}$

$$B_n \equiv [\bar{n} \ n] = \alpha^{-n} + \alpha^n \quad (3)$$

B_0

B_1

B_2

B_3

2

$\alpha^{-1} + \alpha$

$\alpha^{-2} + \alpha^2$

$\alpha^{-3} + \alpha^3 \dots$

Ex.

$$(\alpha^{-3} + \alpha^3)(\alpha^{-5} + \alpha^5) = \alpha^{-8} + \alpha^{-2} + \alpha^2 + \alpha^8$$

$+ B_8$

$$2 + \alpha^8 = B_0 + B_8$$

Alternating binomials $A_n, n \in \mathbb{Z}$

$$A_n \equiv [\bar{n}^a n] = -\alpha^{-n} + \alpha^n$$

$$A_{\bar{n}} = \overline{A_n}$$

$$A_{-1}$$

$$A_1$$

$$A_2$$

$$-\alpha^{-1} + \alpha$$

$$-\alpha^{-2} + \alpha^2$$

Products of alternating & symmetric

$$A_m B_n = (\alpha^{-m} + \alpha^m)(\alpha^{-n} + \alpha^n) = -\alpha^{-(m+n)} + \alpha^{m-n} - \alpha^{n-m} + \alpha^{m+n}$$

$$A_m + A_{m+n} = -A_{n-m} + A_{n+m}$$

$$B_n = -A_{n-1} + A_{n+1}$$

$$(B_1 + B_2 + \dots + B_k) = -A_{k-1} + A_{k+1}$$

$$(B_1 + B_2 + \dots + B_k) = A_{k+1}$$

$$B_m = A_{m+1}$$

Special

Lemma

$$\alpha^n = \alpha^{-(m+n)} - \alpha^{m-n} - \alpha^{n-m} + \alpha^{m+n}$$

$$B_{m+n}$$

⑤

Exer. $A_m^k = ??$ ♦

Products of alternating & symmetric

$$A_m B_n = (-\alpha^{-m} + \alpha^m)(\alpha^{-n} + \alpha^n) = -\alpha^{-(m+n)} + \alpha^{m-n} - \alpha^{n-m} + \alpha^{m+n}$$

$$A_m B_n = A_{m-n} + A_{m+n} = -A_{n-m} + A_{n+m}$$

special case:

$$A_1 B_n = -A_{n-1} + A_{n+1}$$

⑥

$$A_1 (B_k + B_{k+2} + \dots + B_\ell) = -A_{k-1} + A_{\ell+1}$$

$k=1$:

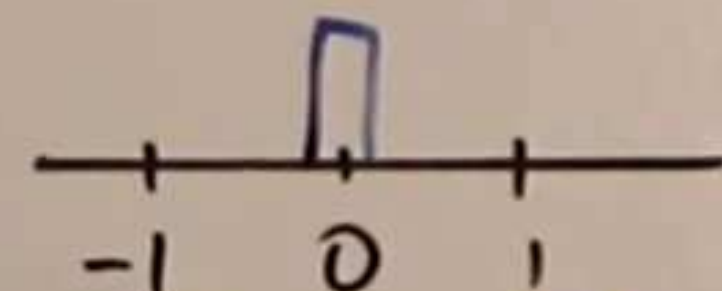
$$(B_1 + B_3 + B_5 + \dots + B_\ell) = A_{\ell+1}$$

$=2$:

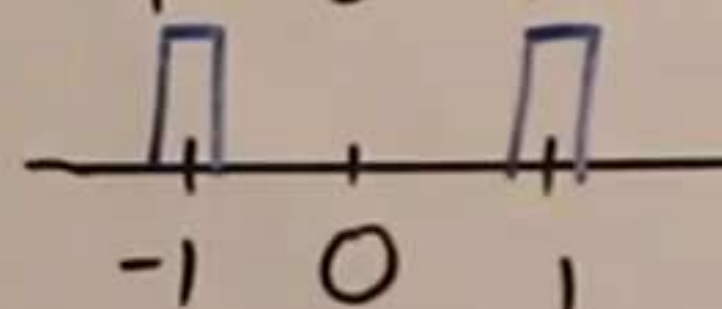
$$(1 + B_2 + B_4 + \dots + B_m) = A_{m+1}$$

Central polynumbers $C_n, n=0,1,2,\dots$

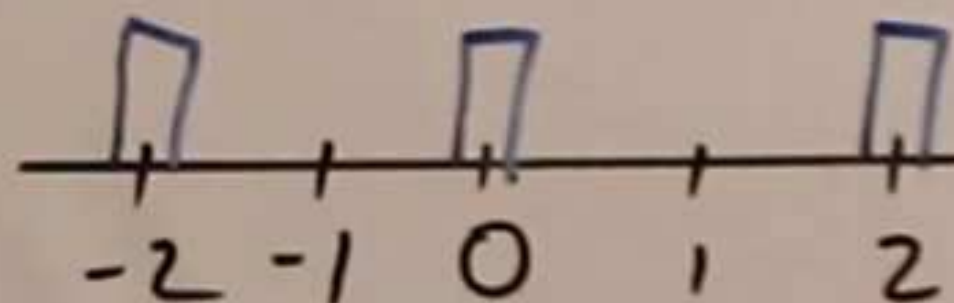
$$C_0 \equiv 1$$



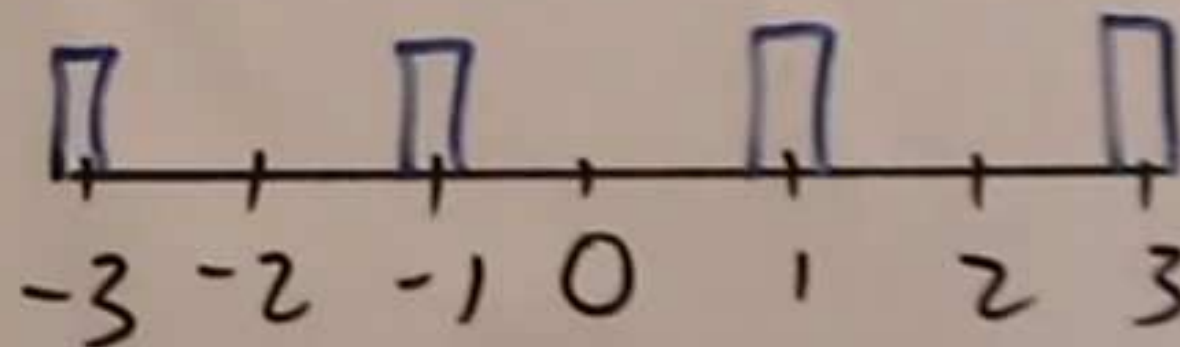
$$C_1 \equiv \alpha^{-1} + \alpha = B_1$$



$$C_2 \equiv \alpha^{-2} + 1 + \alpha^2 = 1 + B_2$$



$$C_3 \equiv \alpha^{-3} + \alpha^{-1} + \alpha + \alpha^3 = B_1 + B_3$$



$$C_4 \equiv \alpha^{-4} + \alpha^{-2} + 1 + \alpha^2 + \alpha^4 = 1 + B_2 + B_4$$

$$C_5 \equiv \alpha^{-5} + \alpha^{-3} + \alpha^{-1} + \alpha + \alpha^3 + \alpha^5 = B_1 + B_3 + B_5$$

⋮

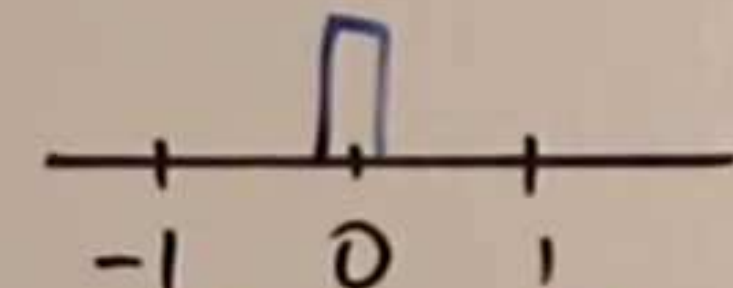
(7)

Central polynomials $C_n, n=0,1,2,\dots$

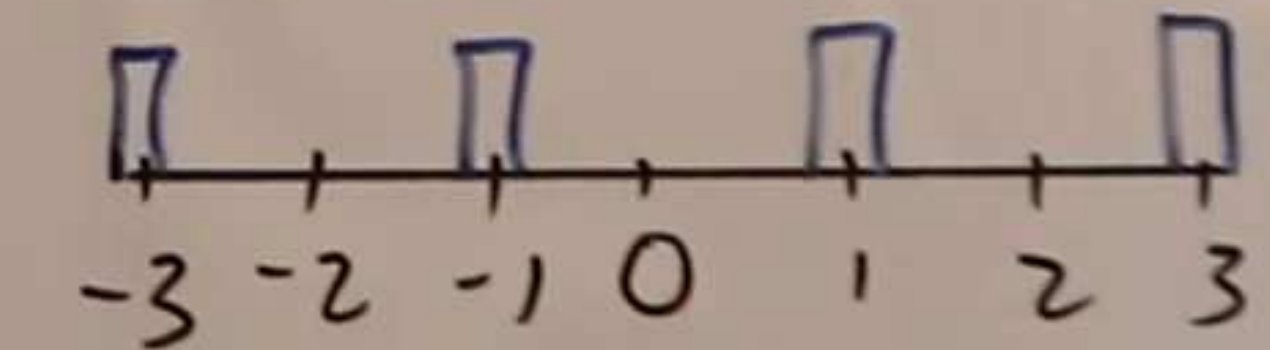
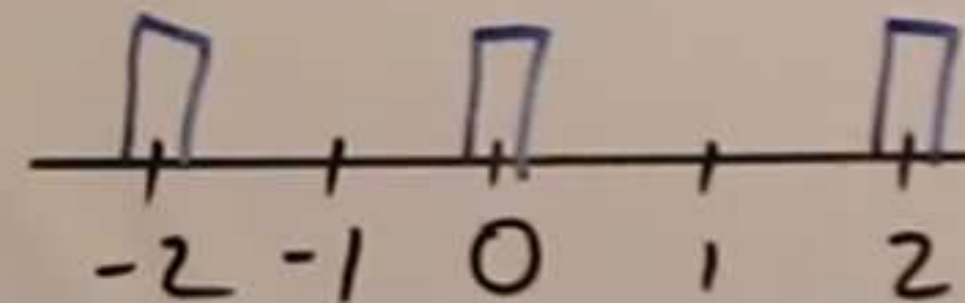
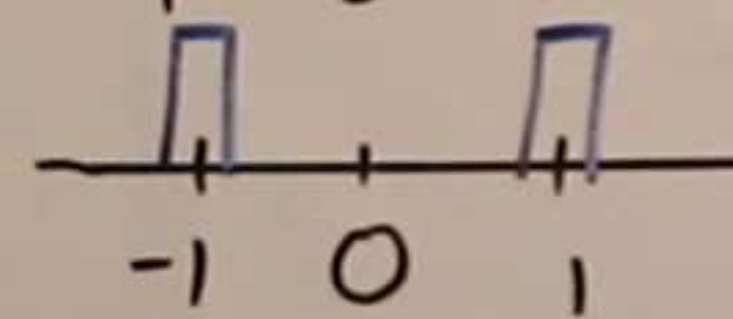
$$C_0 \equiv 1$$

$$\alpha^{-1} + \alpha = B_1$$

$$B_2$$



(7)



$$+B_3$$

$$1+B_2+B_4$$

$$\alpha + \alpha^5 = B_1 + B_3 + B_5$$

The central importance of central polynomials

angular momentum
q-series

Gaussian
binomial
coefficients

Chebyshev
polynomials

$(C_0, C_1, C_2, C_3, C_4, \dots)$

random walks

electroweak
force (physics)

hypergroups

acter

(8)

Next in the series:

MathFoundations235:

**Central polynumbers
and $SU(2)$ characters**